

Statistics for International Commerce

Week 15: Time Series and Forecasting

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Agenda

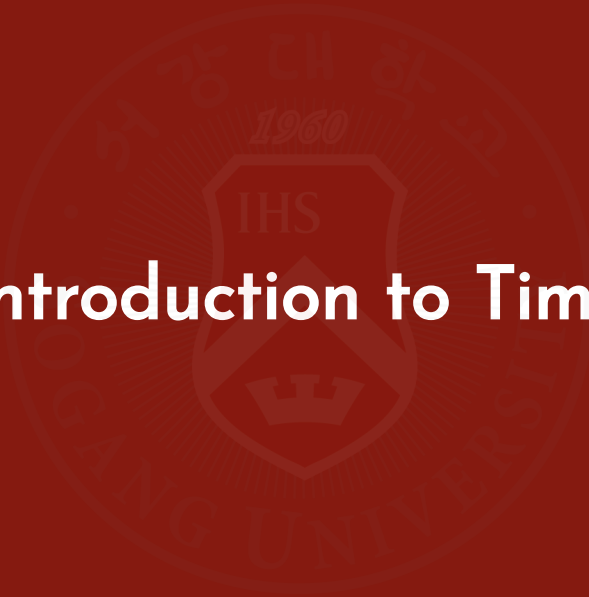
- Part 1. Introduction to Time Series
- Part 2. Components of Time Series
- Part 3. Forecasting Methods
- Part 4. Forecast Accuracy
- Part 5. Python Example
- Part 6. Business Case



Learning Objectives

After this lecture, you should be able to:

- explain what a time series is and distinguish it from cross-sectional data
- identify and interpret trend, seasonal, cyclical, and random components
- compute forecasts using naive, moving average, weighted moving average, and exponential smoothing methods
- evaluate and compare forecast performance using MAD and MSE
- implement and visualize a simple moving-average forecast in Python
- use forecasting results to support basic inventory planning decisions

The background of the slide features a large, faint watermark of the Seoul National University (SNU) logo. The logo is circular, with the university's name in Korean and English around the perimeter. In the center, there is a shield with the letters 'IHS' and a crown below it. The year '1960' is also visible within the logo.

Part 1. Introduction to Time Series

What Is a Time Series?

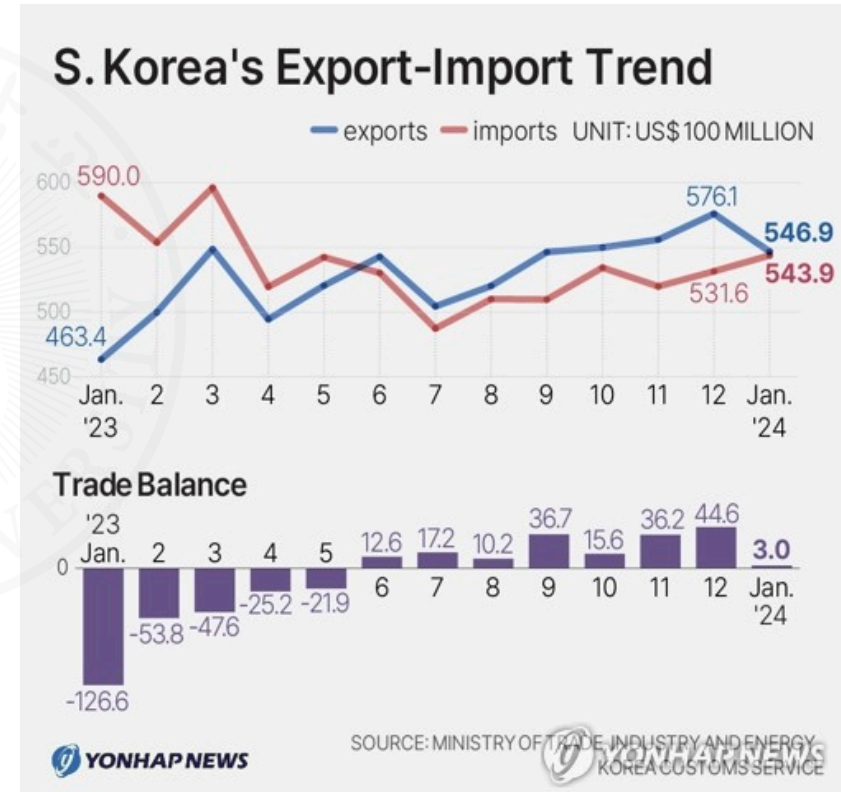
A time series is a sequence of observations collected over time.

Examples:

- monthly exports of Korea
- daily exchange rates
- quarterly GDP
- monthly company sales
- weekly website visitors

Question:

Why is time important here?



Cross-Sectional vs Time Series Data

Cross-sectional	Time series
Many units	One unit
One point in time	Multiple time periods
Countries in 2025	Korea from 2010–2025

Examples:

Cross-sectional:

- GDP of 50 countries in 2025

Time series:

- Korea's GDP from 2000–2025

Illustrative Example: Why Order Matters

Suppose weekly online sales are:

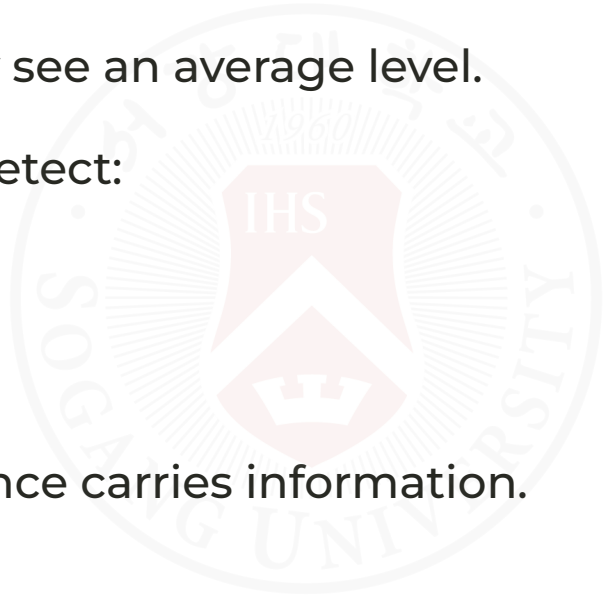
120, 90, 140, 100

If we ignore time order, we only see an average level.

If we keep time order, we can detect:

- ups and downs
- possible trend
- possible seasonal pattern

Key point: in time series, sequence carries information.

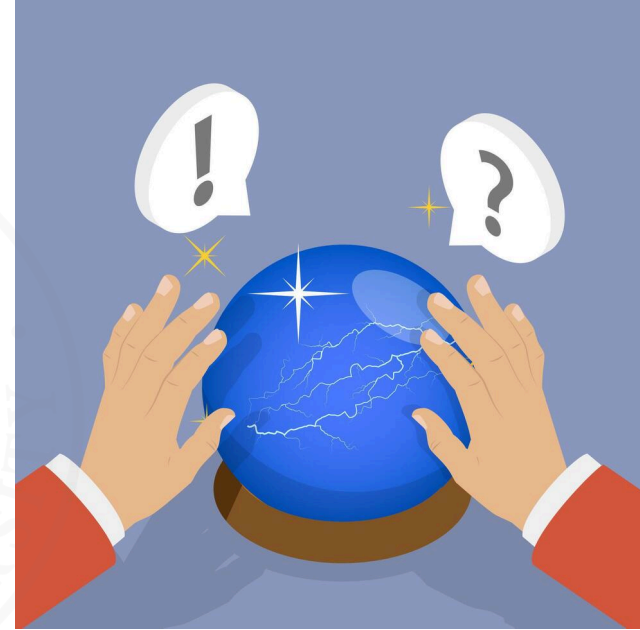


Why Forecast?

Businesses use forecasts for:

- inventory planning
- staffing
- budgeting
- logistics
- production planning
- investment decisions

| To reduce uncertainty and improve decision-making.

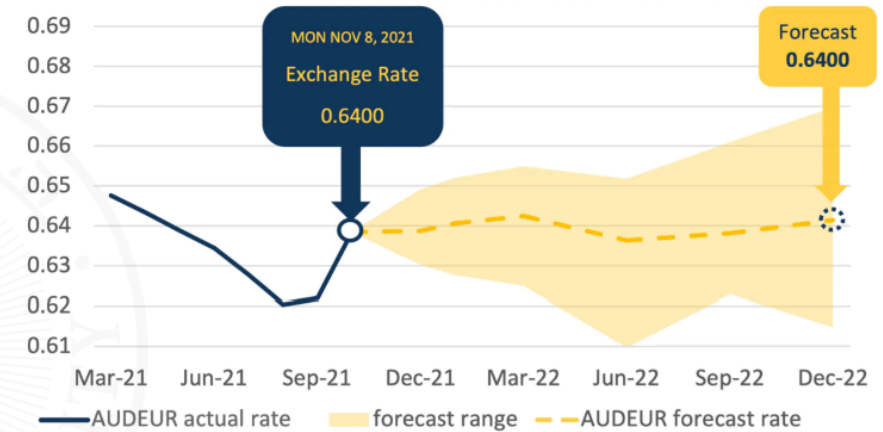


Forecast Examples

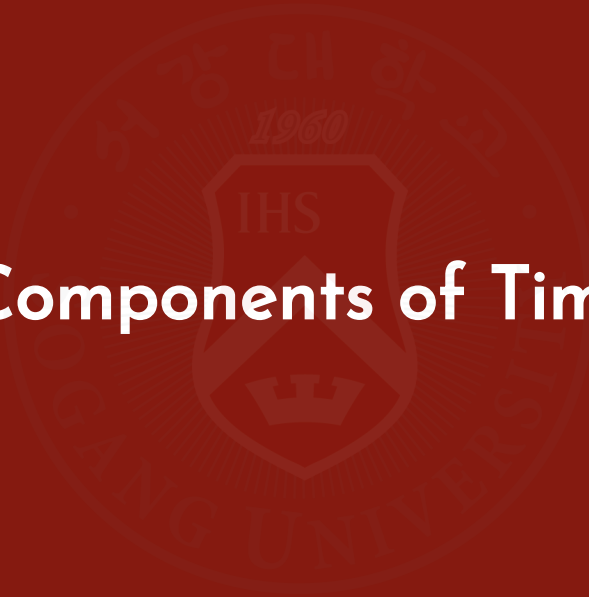
Forecast:

- exports
- imports
- shipping demand
- exchange rates
- tourism arrivals
- online sales

AUDEUR forecasts from bank majors 2023



Source: Multinational bank aggregate data

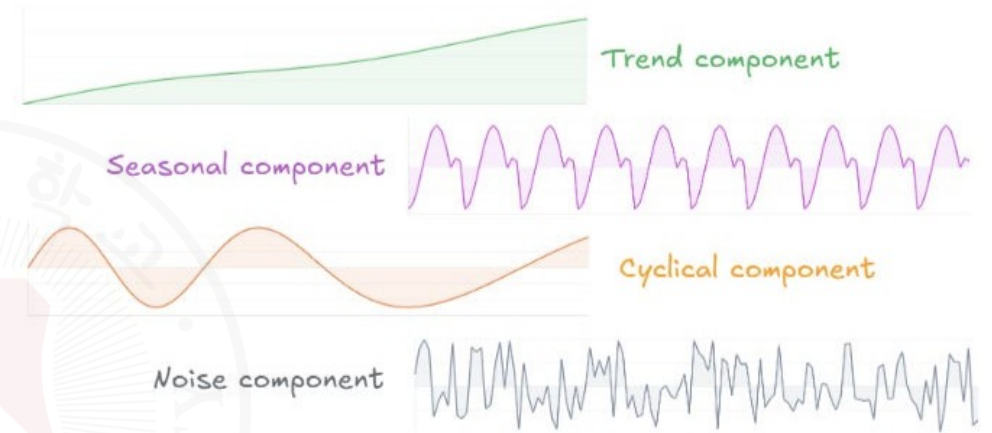
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Part 2. Components of Time Series

Four Components of Time Series

A time series may contain:

1. Trend
2. Seasonal variation
3. Cyclical variation
4. Random variation

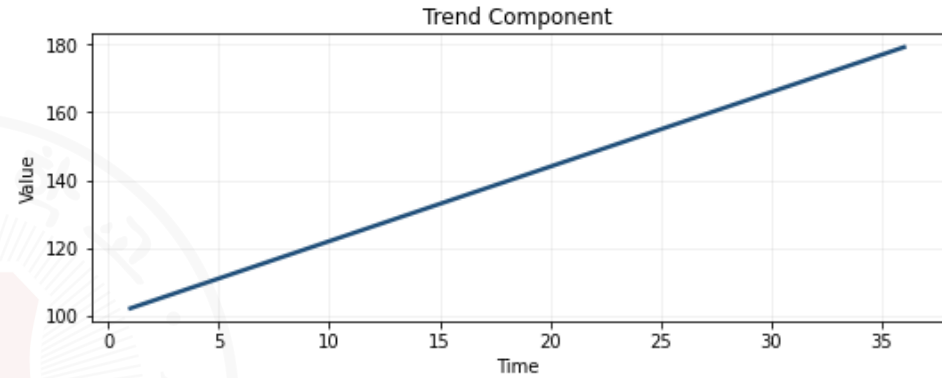


Trend

Long-run movement.

Examples:

- growth in exports
- growth in e-commerce sales



Key point: trend shows the overall direction of the series over time.

Seasonality

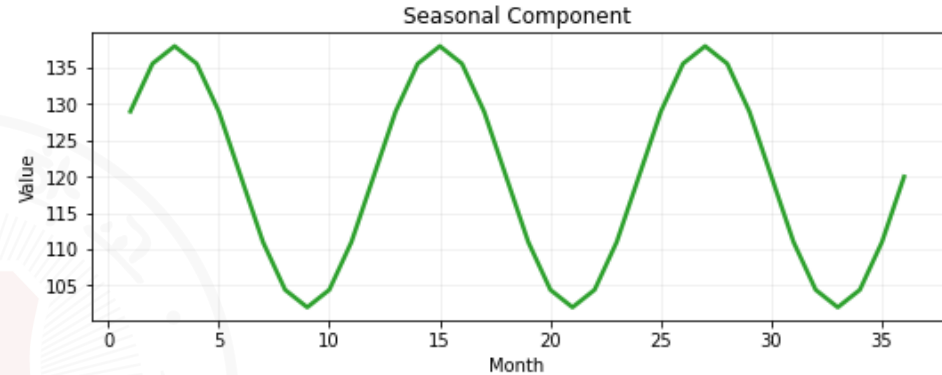
Pattern repeating regularly.

Examples:

- tourism demand
- holiday sales
- airline passengers

Example:

December sales often exceed February sales.



Key point: seasonality reflects regular fluctuations tied to calendar periods.

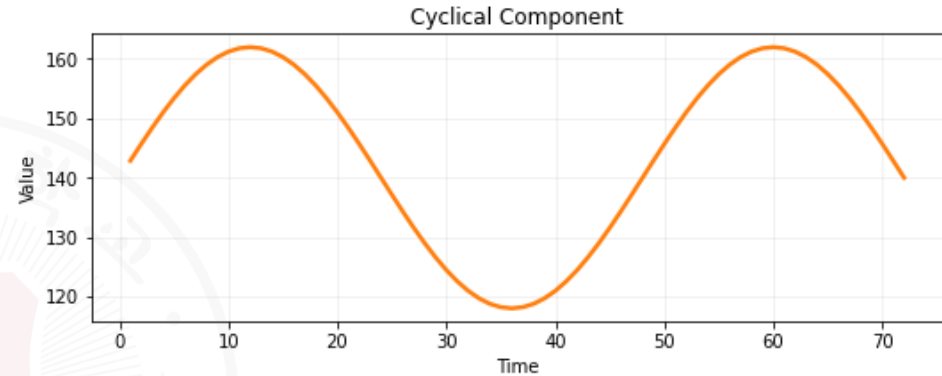
Cyclical Variation

Long-term business cycle effects.

Examples:

- recessions
- booms
- financial crises

Key point: cyclical variation reflects economic expansions and contractions over several years.



Random Variation

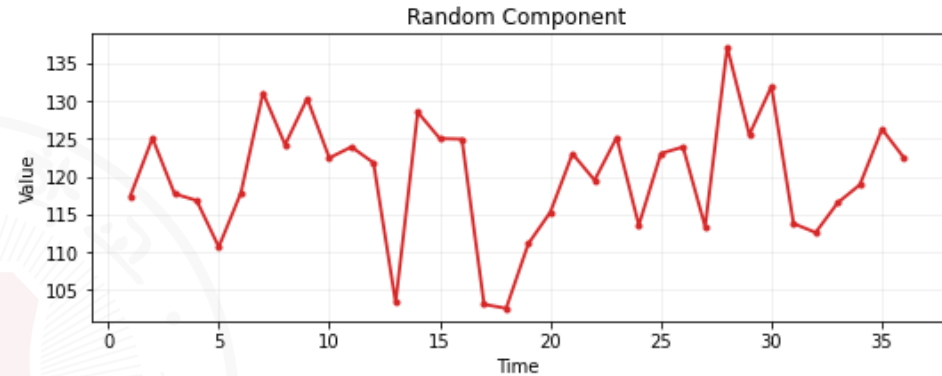
Unexpected events.

Examples:

- natural disasters
- strikes
- pandemics

Cannot be predicted perfectly.

Key point: random variation represents unpredictable fluctuations that cannot be systematically explained.



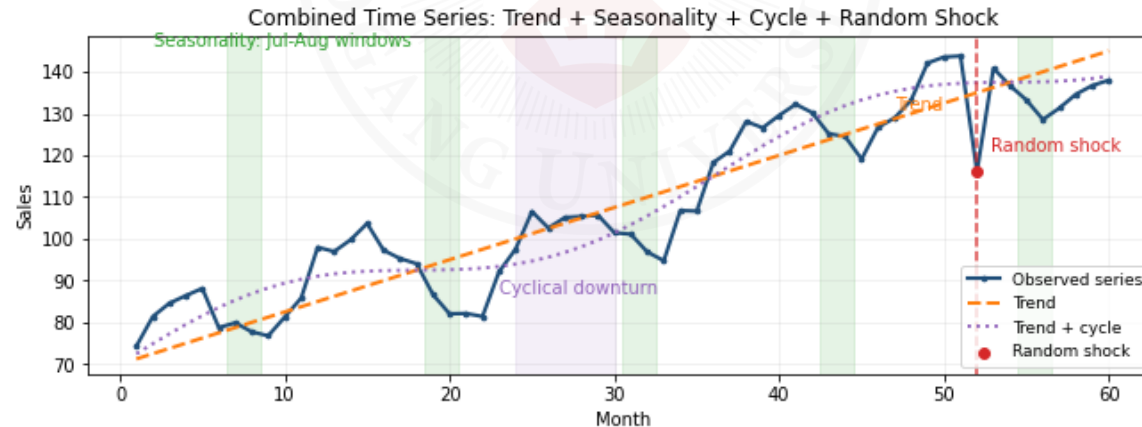
Quick Component Identification Example

Monthly ice cream sales show:

- upward movement over 5 years
- peaks every July-August
- decline during recession years
- unusual drop during a pandemic month

Interpretation:

1. upward movement -> trend (because of growing demand)
2. July-August peaks -> seasonality (because of summer)
3. recession decline -> cyclical variation (because of economic cycle)
4. pandemic drop -> random variation (because of unexpected shock)



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Part 3. Forecasting Methods

Naïve Forecast

Forecast next period using current period.

$$F_{t+1} = Y_t$$

Example:

Current month's sales:

[120]

Forecast next month:

[120]

| Key point: naïve forecast is simple but ignores trends and seasonality.

Moving Average

Forecast uses several previous observations, where 'moving' refers to the window of past observations considered.

Example:

Sales:

110, 120, 130, 140, 170, ...

3-period moving average:

$$\frac{110 + 120 + 130}{3} = 120(\text{for month 4}); \frac{120 + 130 + 140}{3} = 130(\text{for month 5})$$

Key point: moving average smooths out short-term fluctuations but may lag behind trends.

Practice

Monthly exports:

80, 90, 60, 110, 75, ...

Compute 3-month moving average forecasts for months 4, 5, and 6.

Answer:

$$(80 + 90 + 60)/3 = 76.67$$

$$(90 + 60 + 110)/3 = 86.67$$

$$(60 + 110 + 75)/3 = 81.67$$

Why Moving Average?

Advantages:

- simple
- smooths fluctuations

Disadvantages:

- reacts slowly to sudden changes

Weighted Moving Average

Recent observations receive higher weights.

Example:

Weights:

0.5, 0.3, 0.2

Observations:

130, 120, 110

Forecast:

$$0.5(130) + 0.3(120) + 0.2(110) = 123$$

Key point: weighted moving average allows more flexibility by emphasizing recent data.

Exponential Smoothing

Most popular business forecasting method.

Formula:

$$F_{t+1} = \alpha Y_t + (1 - \alpha)F_t$$

where:

$0 < \alpha < 1$ (smoothing constant, indicates weight on current observation)

Key point: exponential smoothing gives more weight to recent observations while still considering past forecasts.

Interpretation of Alpha

Large α responds quickly; small α gives smoother forecasts.

Example

$\alpha = 0.30$ (moderate weight on current observation)

Current sales:

[100]

Previous forecast:

[90]

Forecast:

$$0.30(100) + 0.70(90) = 93$$

Illustrative Example: Choosing Alpha

Assume:

- current actual demand $Y_t = 120$
- previous forecast $F_t = 100$

If $\alpha = 0.8$:

$$F_{t+1} = 0.8(120) + 0.2(100) = 116$$

If $\alpha = 0.2$:

$$F_{t+1} = 0.2(120) + 0.8(100) = 104$$

Higher α reacts faster to new information.

The background of the slide features a large, faint, circular seal of Gangneung University. The seal contains the university's name in Korean and English, the year 1960, and a central shield with the letters 'IHS' and a crown.

Part 4. Forecast Accuracy

Forecast Errors

Forecast error:

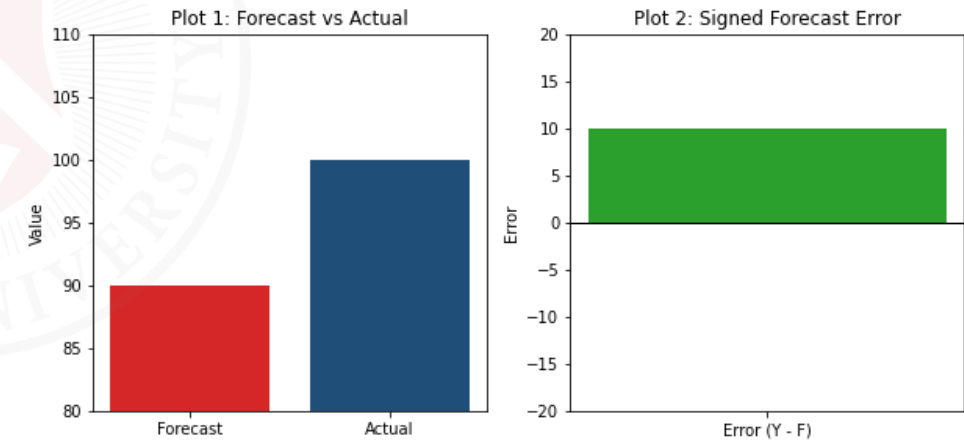
$$e_t = Y_t - F_t$$

Example:

- Actual: 100
- Forecast: 90
- Error: 10

(80.0, 110.0)

(-20.0, 20.0)



Positive error means under-forecasting; negative error means over-forecasting.

Mean Absolute Deviation (MAD)

$$\text{MAD} = \frac{\sum |e_t|}{n}$$

Measures average forecasting error.

Smaller is better.

| Key point: MAD treats all errors equally regardless of direction.

Mean Squared Error (MSE)

$$\text{MSE} = \frac{\sum e_t^2}{n}$$

Large errors receive heavier penalties.

| Key point: MSE emphasizes larger errors more than MAD.

Choose the method with smaller MAD and smaller MSE.

Worked Accuracy Example

Actual demand for 4 months: 100, 110, 120, 130

Method A

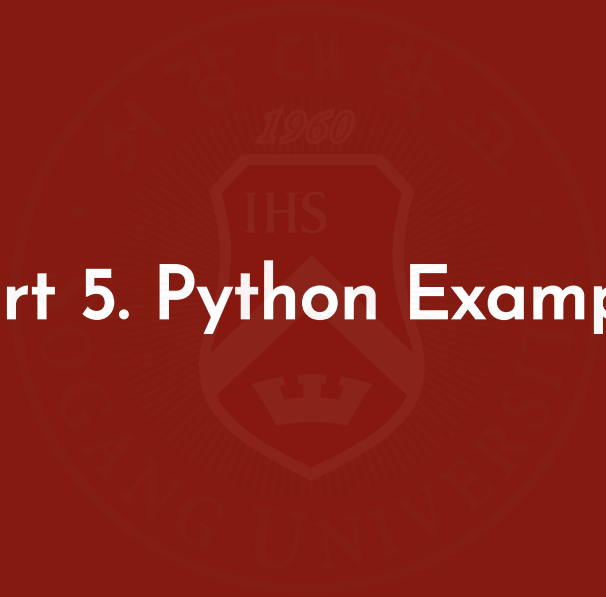
- Forecast: 98, 112, 118, 128
- Errors: 2, 2, 2, 2
- MAD = 2.0
- MSE = 4.0

Method B

- Forecast: 90, 105, 130, 125
- Errors: 10, 5, 10, 5
- MAD = 7.5
- MSE = 62.5

Conclusion: Method A is more accurate because it has the smaller MAD and MSE.

Part 5. Python Example



Why Python Here?

Python helps us turn the lecture ideas into a small forecasting workflow:

- build a dataset
- visualize the series
- create forecasts
- compare forecast accuracy

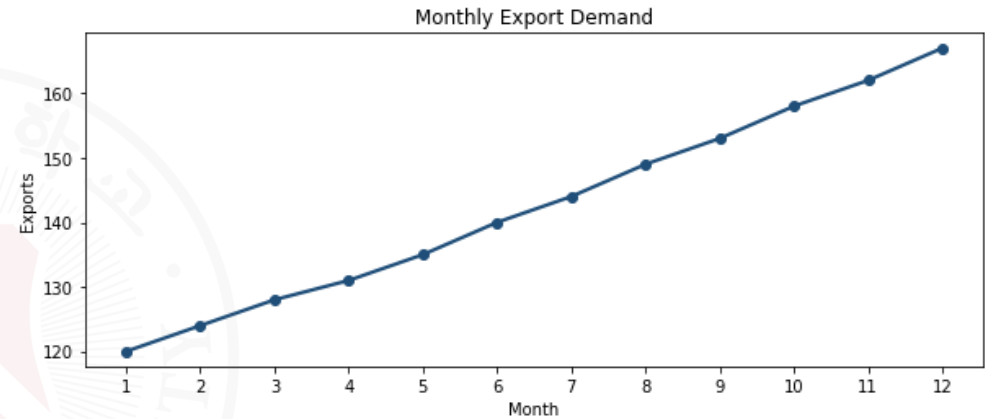


Example Dataset and Trend

Monthly export demand shows a steady upward pattern.

Month	Exports
1	120
2	124
3	128
4	131
5	135
6	140
7	144
8	149
9	153
10	158
11	162
12	167

```
## ([<matplotlib.axis.XTick object at 0x000002116AA9AE90>, <matplotlib.axis.XTick
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Interpretation:

- the level is rising over time
- month-to-month changes are small
- this is a good setting for comparing simple forecasting methods

The line slopes upward, which is why simple forecasting methods are useful here.

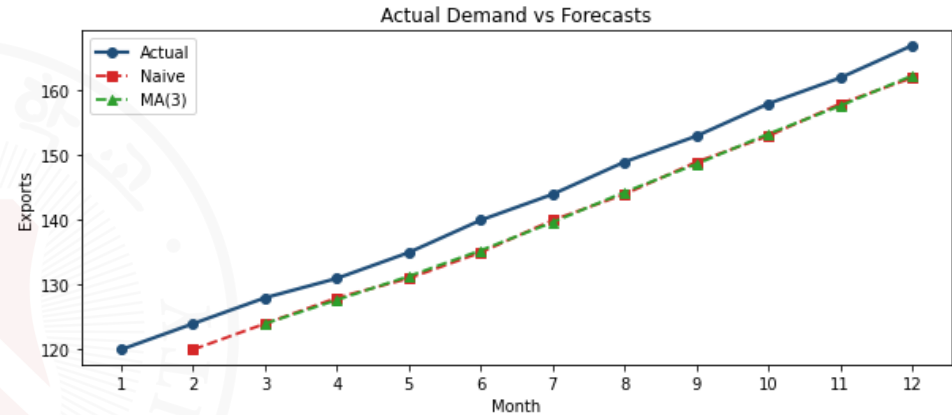
Forecast Output

The Python workflow produces two forecasts:

Month	Actual	Naive	MA(3)
4	131	128.0	124.0
5	135	131.0	127.7
6	140	135.0	131.3
7	144	140.0	135.3
8	149	144.0	139.7
9	153	149.0	144.3
10	158	153.0	148.7
11	162	158.0	153.3
12	167	162.0	157.7

The chart makes the lag visible at a glance.

```
## ([<matplotlib.axis.XTick object at 0x000002116AB7DD10>, <matplotlib.axis.XTic
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Interpretation:

- the naive forecast follows the most recent value
- MA(3) is smoother, but it lags behind the rising trend

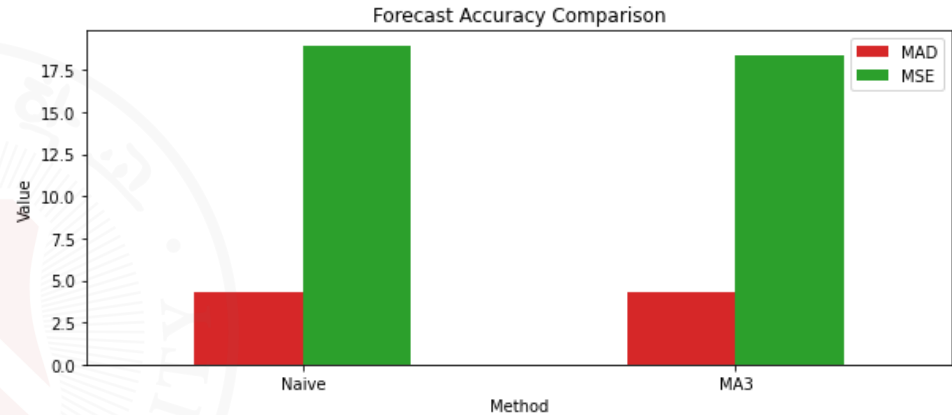
Accuracy Summary

Method	MAD	MSE
Naive	4.27	18.64
MA(3)	8.56	73.82

```
## (array([0, 1]), [Text(0, 0, 'Naive'), Text(1, 0, 'MA3')])
```

Conclusion:

- for this steadily rising series, the naive forecast is more accurate numerically
- MA(3) is still useful because it smooths short-term fluctuations
- MAD and MSE make the comparison objective and easy to explain

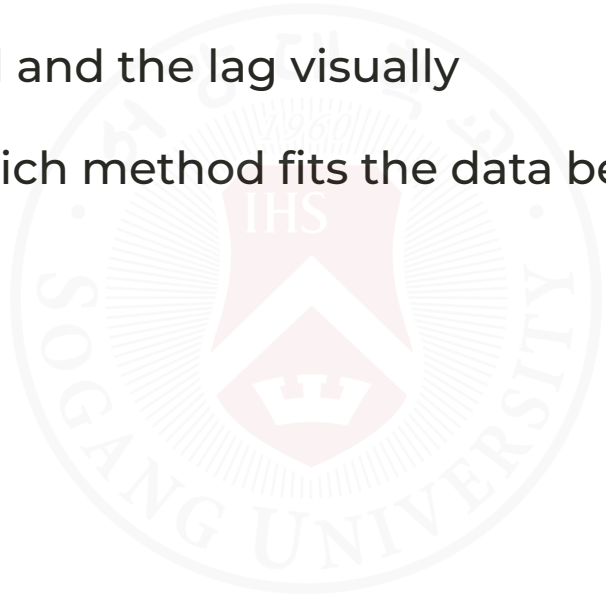


Lower bars are better, and both MAD and MSE favor the naive forecast here.

What to Remember

The key lesson is not only how to code the forecast, but how to read the output:

- tables show the forecast values directly
- figures would show the trend and the lag visually
- accuracy measures tell us which method fits the data better



The seal of Gangneung University is a circular emblem. It features a central shield with a crown on top and the letters 'IHS' inside. Above the shield is the year '1960'. The outer ring of the seal contains the university's name in Korean '강릉대학교' at the top and 'GANGNEUNG UNIVERSITY' at the bottom.

Part 6. Business Case

Case Study

A Korean exporter observes monthly export demand:

Month	Orders
Jan	120
Feb	130
Mar	140
Apr	145
May	160

Tasks:

1. Calculate 3-month moving average.
2. Forecast June demand.
3. Recommend inventory planning.

What We Learned

Time series components:

- trend
- seasonality
- cycles
- randomness

Forecasting methods:

- naïve
- moving average
- weighted moving average
- exponential smoothing

Forecast evaluation:

- MAD
- MSE



Final Message

Forecasts are never perfect.

Good forecasting reduces uncertainty and improves business decision-making.



Remaining Schedule

(June 9) Multiple Time Series and Forecasting (LMW Chapter 16)

- Final review and Q&A session

(June 11) Project presentations

(June 16) No class (final exam week)

(June 18) Final Exam



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Any questions?

Thank you for your attention and active participation!